TUTORIAL : HOW TO EDIT THE GAME'S CODE (and UI text) (and probably more stuff I assume)

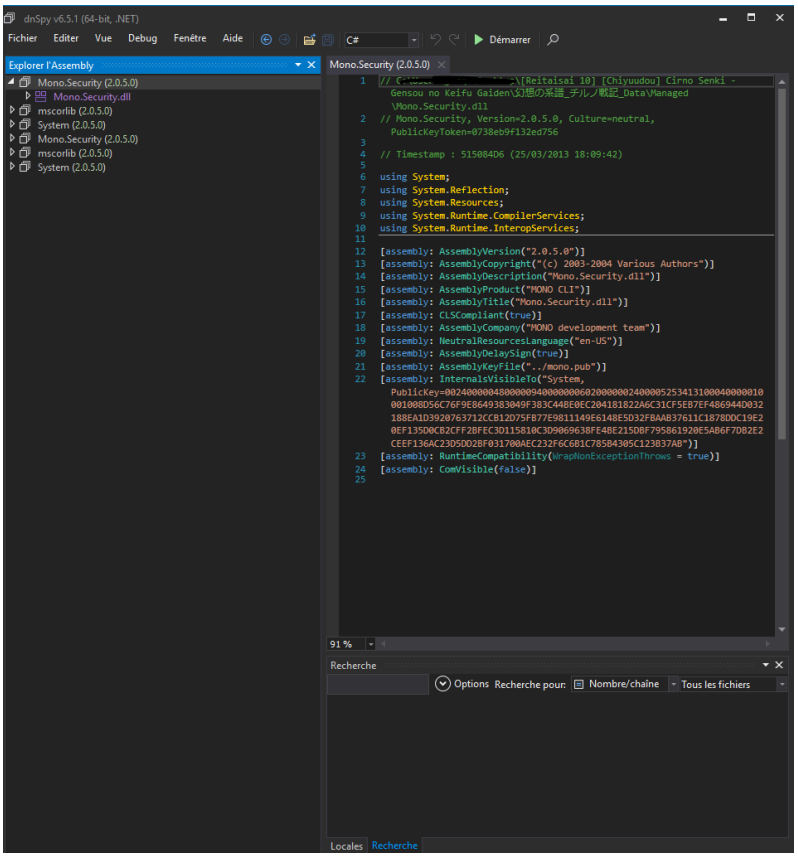
Hey, it's Gigafraud ! I'll explain to you how to edit the game's code itself !

Just for a disclaimer, if you plan on actually changing the code, you probably need a little bit of knowledge on programming.
If you just want to edit the UI's text, it's not necessary.

Alright, let's get started !

We'll be using a software called dnSpy : https://github.com/dnSpy/dnSpy
This will allow us to do pretty much anything we want to do ! (if you know what you're doing, of course)

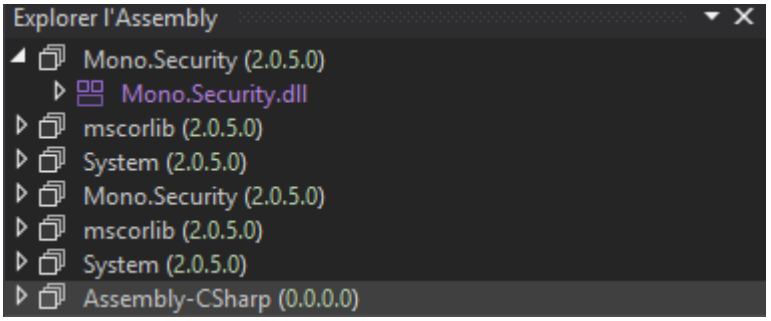
The UI looks like this :



(bad screenshot i know)

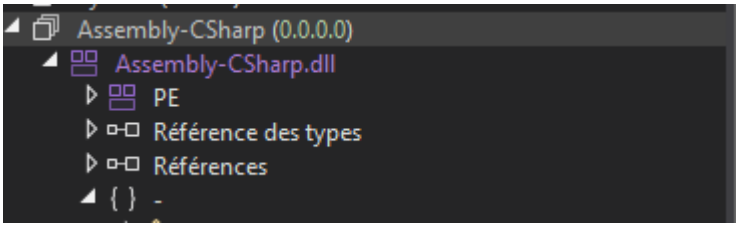
The left part contains the files and the classes you will edit, the upper left contains the code of said class, and the bottom right will allow us to look for characters in the file.

First, we'll go to File/Open, and open the following file :
幻想の系譜_Data/Managed/Assembly-CSharp.dll

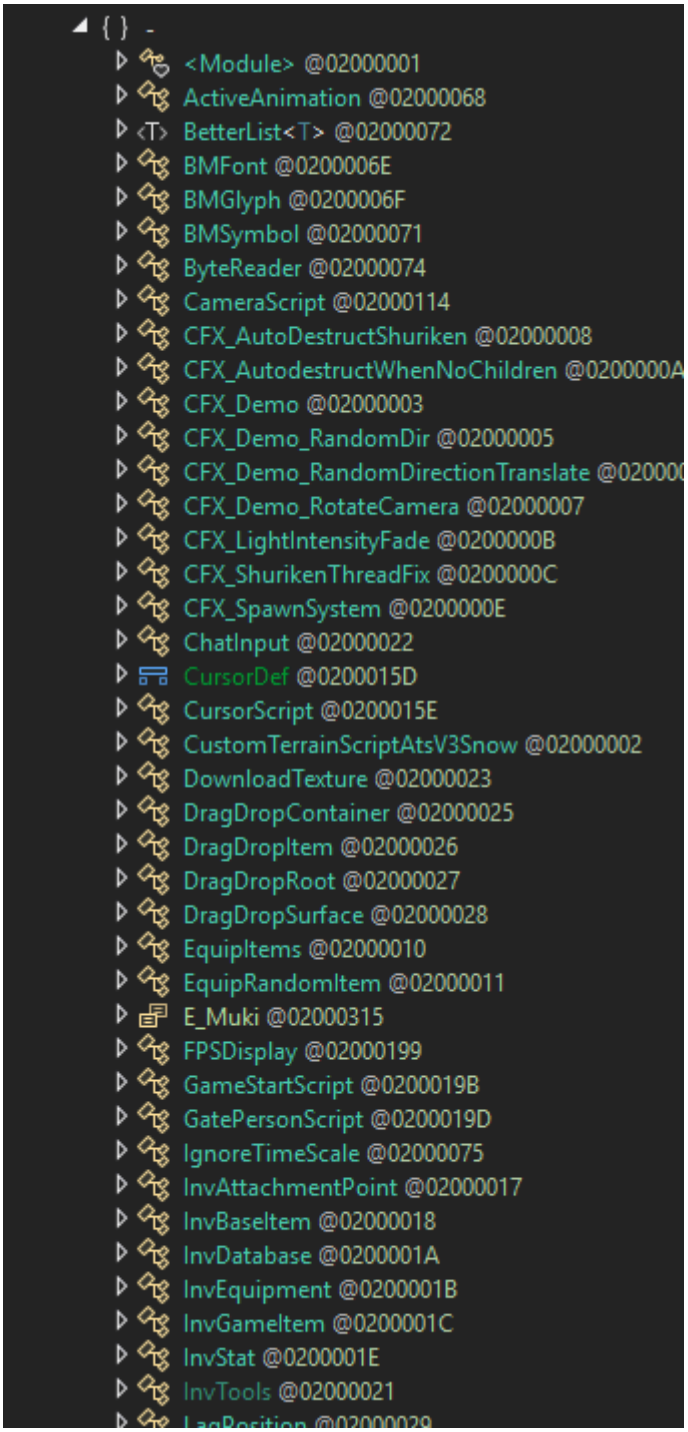


It now appears on the left.

Now click on the arrow on the left, then do it again, then the arrow on the left of this option :
> { } -



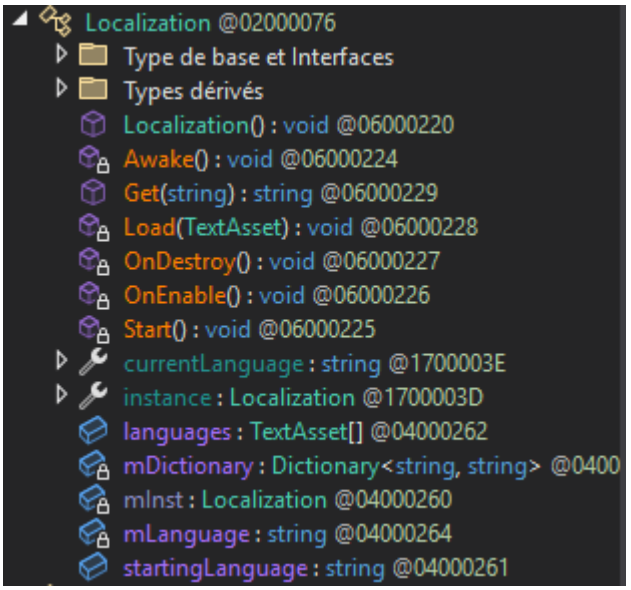
And suddenly :



Bam ! A lot of classes are here.

The program is in C#. if you know C, C++ or even Java, the syntax will look familiar (I'm sorry if you only know Python)

By clicking on the arrows next to the classes, it will show you the attributes and the functions/methods of this class.
For example :



But you'll be telling me that you don't care about any of these classes ! Where are the items or the characters ?
It's actually in another location :
Now scroll down and click on this arrow :

In this, you'll see way more interesting classes, like Item for example.
Here are some interesting classes you can look at :
-Yusoutai (the convoy)
-Item, ItemPerson
-Junbi (the menu before starting a map)
-Person
…

- MessageDisp @020001D1
- MikirareSkill @020000F1
- MojiEffect @02000172
- MojiEffectForSkill @02000173
- Nanido @02000305
- NGUI2D @020002F7
- NPC @02000303
- Option @02000308
- OverYusoutai @0200040D
- Person @0200030D
- PrefabEffect @02000181
- ReClassChangeAction @020001C3
- SaidouNoAnime @02000115
- SaudouTurn @02000116
- SaveCommandBox @02000325
- SaveCommandItem @02000324
- SaveData @02000326
- SaveLoad @02000327
- ScriptState @020001D0
- Senseki @02000329
- SensekiPerson @02000328
- Sentakushi @02000145
- Setting1 @020001A0
- Setting2 @020001A1
- Setting3 @020001A2
- Setting3Save @020001A3
- Shien @0200032A
- ShienKaiwaChoice @0200032D
- Shop @02000332
- ShopData @02000333
- ShopItemCommandItem @0200012B
- Skill @020001B3
- SkillDisp @02000335
- SkillDispMikirare @02000337
- SkillEffect @02000174
- SkillMikirareEffect @02000175
- SkinClass @02000339
- StatusDisp @020003B3
- StatusHistory @0200030C
- StatusUp @020001C2
- TalkFlg @02000183
- TatiePlace @020001CF
- TenseiChange @020001BF
- TextureItem @020002F5
- Title @020003B4
- TurnEnd @020003B9
- TurnStart @020003BA
- UnitChoice @020003C5
- UnitCommandBox @020003D4
- UnitItem @020003E4
- UnitJunbiBase @020003C4

How cool is that ?

Now if you want to edit the functions, you right-click on the code, then on Edit Class (C#).

It'll open this window :

odifier la classe - Person @0200030D

```
12      {
13          // Token: 0x060014CA RID: 5322 RVA: 0x0006ABD8 File Offset: 0x00068DD8
14          public Person()
15          {
16          }
17
18          // Token: 0x060014CB RID: 5323 RVA: 0x0000FC82 File Offset: 0x0000DE82
19          static Person()
20          {
21          }
22
23          // Token: 0x060014CC RID: 5324 RVA: 0x0006AC7C File Offset: 0x00068E7C
24          public void SetLevelData(DataContent pData, string sLevel)
25          {
26              this.MaxHP = int.Parse(pData.GetData(sLevel, "体力"));
27              this.HP = this.MaxHP;
28              this.TargetHP = this.HP;
29              this.ChikaraOriginal = int.Parse(pData.GetData(sLevel, "力"));
30              this.MaryokuOriginal = int.Parse(pData.GetData(sLevel, "魔力"));
31              this.WazaOriginal = int.Parse(pData.GetData(sLevel, "技"));
32              this.HayasaOriginal = int.Parse(pData.GetData(sLevel, "速さ"));
33              this.KouunOriginal = int.Parse(pData.GetData(sLevel, "幸運"));
34              this.ShubiOriginal = int.Parse(pData.GetData(sLevel, "守備"));
35              this.MabouOriginal = int.Parse(pData.GetData(sLevel, "魔防"));
36              this.OriginalIdou = int.Parse(pData.GetData(sLevel, "移動"));
37              this.Taikaku = int.Parse(pData.GetData(sLevel, "体格"));
38              string data = pData.GetData(sLevel, "索敵");
39              if (!string.IsNullOrEmpty(data))
40              {
41                  this.SakutekiRangeBase = int.Parse(data);
42              }
43              else
44              {
45                  this.SakutekiRangeBase = 3;
46              }
47              this.SkillList = pData.GetData(sLevel, "スキル", true);
48              this.SkillPotentialList = this.SkillList.Where((string s) => s.StartsWith("*")).ToList<string>();
49              this.SkillList.RemoveAll((string s) => this.SkillPotentialList.Contains(s) || string.IsNullOrEmpty(s));
50              this.SkillPotentialList = this.SkillPotentialList.Select((string s) => s.Substring(1)).ToList<string>();
51              this.KousokuLevel = pData.GetData(sLevel, "速");
52              this.KousokuLevel = ((!(this.KousokuLevel == string.Empty)) ? this.KousokuLevel.Trim() : "--");
```

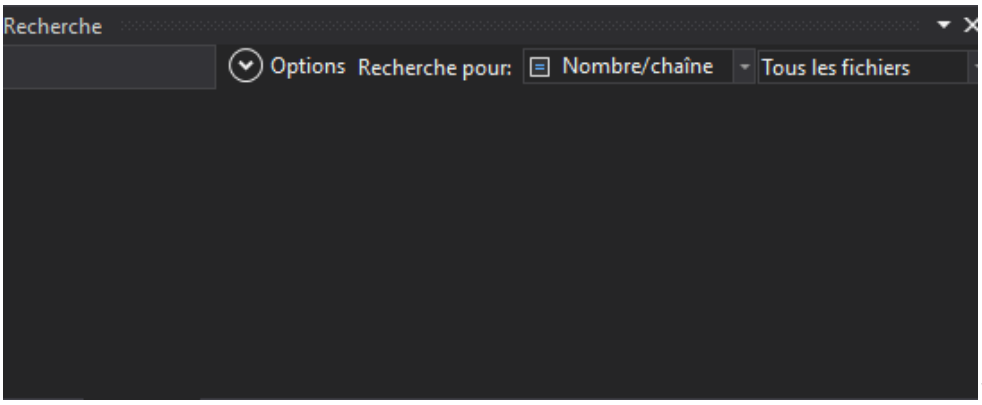
Here, you can edit the code the way you want. When you're done, click on Compile on the bottom right. As long as you didn't destroy everything, it should compile correctly. If you can't fix what you did, just close the window without saving, or use CRLT+Z.

Sometimes it fails to compile even if you didn't edit anything, don't ask me why.
When you're done, go to File/save all… and it will save the new .dll file. Try to make backups in order to not destroy your game.

A very powerful tool !

From now on, I'll be talking specifically to the people that want to translate either Cirno Senki or Hifuu Bouenkyou.

You can use the bottom-right part of the UI to search for specific strings :



yes it's in french

Switch the option to Number/Strings to find the strings you want to edit.

One way to do it easily is :

1) Play the game and get to a japanese text you want to edit.
2) Take a screenshot of this text, then use an image to text tool to get the japanese characters.
3) Paste these characters in the search bar, and look at the files containing this text

If you want to be sure about what to do, you can download Gensou No Keifu ~ Hajimari No Miko and open the AssemblyCSharp.dll of this game. Here, everything is translated well (courtesy of Bob Bob), so you can compare the files from Keifu to the ones you edit. Now of course there will be differences (especially if you are trying to edit Bouenkyou), but it's still useful.

!!! BUG !!!

But I'll talk about a little problem in the original code.

If you tried to edit the weapons' display name thanks to AssetBundleExtractor, you might have noticed that the weapons lose their "special effects". For example, Freeze weapons cannot inflict frostbite, or poison weapons cannot inflict Poison, or even staves cannot heal anymore.

The problem is : **the class that gives these effects to weapons looks for the weapon's display name instead of the system name. It should instead look for the system name.**

It's not a problem when it's in japanese, since the system name and the display name is the same.

So if you change the display name, this class doesn't recognize the weapon and won't give its effect. To remedy this, you need to edit the function that calls this, and change DispName to Name.

Finally, I'll talk about UI elements, like menus and such.
Here's one of them :

```
public class Yusoutai
{
    // Token: 0x06001B3A RID: 6970 RVA: 0x00094208 File Offset: 0x00092408
    public Yusoutai(PersonScript p)
    {
        this.m_canCancel = true;
        this.m_tYusouTaidata = GameStartScript.GD.YusouData;
        this.m_pPerson = p;
        float num = 100f;
        float num2 = (float)(Screen.height / 2 - 150);
        this.nameRect = new Rect(num, num2, 250f, 30f);
        this.personItemBox = new Rect(num, num2 + 50f, 250f, 205f);
        this.yusouTitleRect = new Rect(num + 260f - 275f, num2 + 250f, 700f, 50f);
        this.yusouItemRect = new Rect(num + 250f + 30f, num2 + 50f, 300f, 205f);
        this.personItemSelectedRect = new Rect(num + 100f, this.yusouItemRect.yMax + 40f, 100f, 105f);
        this.yusouItemSelectedRect = new Rect(this.personItemSelectedRect);
        this.yusouItemSelectedRect.x = this.yusouItemSelectedRect.x + 150f;
        this.yusouItemSelectedRect.width = 130f;
        this.SetPersonBox();
        List<string> list = new List<string> { "All", "Fast", "Homing", "Large", "Charms", "Magic", "Other" };
        this.m_tItemKind = new CommandBox(list.Select((string s) => new CommandItem(s)).ToList<CommandItem>(),
          list.Count, "yusouTitle");
        this.m_tItemKind.DispRect = this.yusouTitleRect;
        this.m_tItemKind.YPadding = 0f;
        this.m_tItemKind.IsActive = 2;
        this.m_tYusouTaidata = GameStartScript.GD.YusouData;
        this.SetYusoutaiBox();
        this.m_tYusouItem.IsActive = 0;
        List<string> list2 = new List<string> { "Recieve", "Trade" };
        this.m_tYusouItemSeletecd = new CommandBox(list2.Select((string s) => new CommandItem(s)).ToList<CommandItem
          (), "personItemSelected");
        this.m_tYusouItemSeletecd.DispRect = this.yusouItemSelectedRect;
        if (this.m_tPersonItem.GetMaxCnt() < 0)
        {
            this.m_tPersonItem.IsActive = 0;
```

Yusoutai

The floats like 100f, 250f and such, are lengths. If you change the values, it will change the size of different things (here, the size of the convoy window).

Here are some stuff you would want to edit :

-Character names
    -Fight preview (Person/EquipItemNameDisp)
    -During fight (DrawEquipXY)

-Weapons (DispName again)
    -Fight preview (Person/EquipItemNameDisp)
    -During fight (DrawEquipXY)
    -Item/ItemKindStr
    -IfMinusThenInf(int, bool)
    -Font sizing (StageScript/DrawPersonOnMap(string, PersonScript, Rect))

-Tile names (StageScript/NormalDraw())
-Menu options
    -Main menu (Title/NGUIupdate)
    -Settings (Option())
    -Between chapter (Junbi(), Junbi/Move())
    -On map
        -On character (StageScript/PersonCommandMove())
        -On empty tile (StageScript/FieldCommandMove())

-Difficulties (Nanido/Draw())
-Chapter names (Attack/DrawAttackDetail)
-Save menu (playtime,...)
-Character summary
    -Stats (StageScript/DrawPersonStatusLeft(ref Rect, PersonScript, bool, bool))
    -Weapon mastery (StageScript/DrawPersonStatusLeft(ref Rect, PersonScript, bool, bool))
    -Defeat count (Senseki/Draw()), StageScript/RetsudenPersonDraw())
-Convoy
    -Window sizes (Yusoutai)
    -Font size (Yusoutai)
    -"No item." (Shop/CommandMove, CommandBoxItemMerge/Draw(),

You still need to edit other stuff of course.

That was everything I know about editing these games. I don’t know if anyone will use these tutorials, but if even one person does, I’ll be happy.

Keep tinkering different stuff with dnSpy to learn things yourself !